Paper 15: Non-Executive World Bible Consistency Monitors

Pillar: Graph-Loop Creative Pipelines — Autonomous Research Ecosystem

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Abstract

We hypothesize that Non-Executive World Bible Consistency Monitors achieves 0 lore contradictions on Python / LangGraph while preserving strict mathematical invariants across edge, creative graph-loop, and decentralized clearing layers. We present a formal mathematical model, a production implementation in `Python / LangGraph`, and an automated empirical verification harness targeting **0 lore contradictions**. All benchmarks and invariants are reproducibly verified via our continuous Research Graph-Loop pipeline [3, 6, 2].

1 Introduction & Formal Hypothesis

Modern autonomous agent architectures face fundamental bottlenecks across constrained edge silicon, deterministic multi-modal synthesis, and decentralized economic clearing. In this work (Paper 15 within the *Graph-Loop Creative Pipelines* pillar), we address these limitations directly.

Core Hypothesis: We hypothesize that Non-Executive World Bible Consistency Monitors achieves 0 lore contradictions on Python / LangGraph while preserving strict mathematical invariants across edge, creative graph-loop, and decentralized clearing layers.

2 Mathematical Formulation

We formulate our system objective and state transition invariant as:

$$\mathcal{L}_{\text{total}}(\theta) = \mathbb{E}_{x\sim\mathcal{D}}\left[\mathcal{D}_{\text{KL}}(P_{\text{target}} \parallel P_{\text{draft}})\right] + \lambda\|\Delta_{\text{state}}\|_2^2 \tag{1}$$

Subject to strict runtime verification bounds (`0 lore contradictions`) executed across our hybrid compute fabric (Local Edge Silicon, Homelab Silo Server, and FAU Research Computing HPC nodes) [4, 5, 1].

3 System Architecture & Implementation

The reference architecture is implemented in `Python / LangGraph` across the following work package modules:

- `python/creative_pipeline/bible_monitor.py`
- `research_ecosystem/pillar2_creative.py`

4 Empirical Evaluation & Benchmarks

We executed the automated verification suite for Paper 15 and recorded the following empirical metrics (Table 1):

Table 1: Empirical Verification Results for Paper 15

Metric / Invariant	Measured Value
caught_violations_count	`3`
bad_route	`script_agent`
good_route	`art_agent`
passed	`True`

5 Conclusion

We have demonstrated that *Non-Executive World Bible Consistency Monitors* satisfies its target specification (`0 lore contradictions`) with deterministic reproducibility.

References

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